

When Underrepresentation Matters: A Framework for Anticipating Fairness Risks

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Underrepresentation of minority groups in training data is widely regarded as a primary driver of unfair model behavior, motivating a broad range of data rebalancing interventions. However, recent work has begun to suggest that whether demographic imbalance actually translates into performance disparities may depend on a variety of dataset properties, which have so far been studied in isolation. In this work, we sketch a methodological pipeline to systematically study the factors that govern the impact of underrepresentation on fairness, with the goal of formulating a unified, empirically testable hypothesis that can be evaluated prior to model training. We additionally outline lightweight strategies for assessing these factors without the need to train a full model, making the approach practical for real-world data preparation workflows.

Keywords: algorithmic fairness, underrepresentation bias, selection bias

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1 Introduction

Machine learning models often exhibit uneven performance across demographic groups, which is frequently attributed to the underrepresentation of disadvantaged groups in training data [11, 22]. The intuition behind this view is that learning algorithms are sensitive to the distribution of data they observe: when some groups appear less frequently, the model has fewer opportunities to capture patterns that are relevant for them, which can result in lower prediction accuracy [2, 3, 10]. This perspective has motivated a broad range of interventions aimed at

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improving dataset representativeness, often by increasing the availability of data for underrepresented groups or by rebalancing training distributions [5, 13, 27].

However, recent evidence suggests that data imbalance alone does not fully explain observed disparities. In some settings, models trained on highly imbalanced datasets maintain performance across groups [9, 19], while in others disparities persist even after rebalancing [4, 14]. These findings indicate that fairness is shaped by multiple interacting factors, and that the consequences of underrepresentation are not fixed but contingent on broader properties of the data and the learning task. Consistent with this view, prior work has proposed hypotheses regarding dataset characteristics that may modulate the impact of underrepresentation. Notably, differences in feature-label relationships across subgroups, referred to as *subgroup predictivity differences*, have been identified as potentially influential [19, 26]: models generalize well under strong imbalance when these relationships are similar across groups, whereas disparities are more likely to emerge when they diverge. Further properties, including label bias [4], features predictivity [16], subgroup separability [15], and task difficulty [6], have each been shown to affect how disparities in representation ultimately translate to performance disparities.

Despite these advances, there is no unified hypothesis that integrates the full range of potentially relevant factors into a coherent account of when underrepresentation meaningfully affects fairness outcomes and warrants intervention. Moreover, there is no established methodological framework for assessing these factors in a lightweight manner before model training, leaving practitioners without actionable guidance at the data preparation stage. To address both gaps, we sketch a methodological pipeline for systematically studying the factors that govern the impact of underrepresentation on fairness, and outline the foundations of a unified, empirically testable hypothesis; additionally, we propose lightweight strategies for measuring these factors without training a full model.

2 Literature Review

The emphasis on underrepresentation as a primary driver of unfairness has long been central to the fairness literature. Early work and survey papers frame representativeness as a key requirement for equitable machine learning, arguing that skewed training distributions result in systematically worse performance for minority groups [3, 20, 23]. This view is reflected in popular accounts of algorithmic bias [7, 21], and formalized in studies that use group prevalence as a proxy for fairness risk [2, 25]. Results from representation learning further support this perspective, showing that imbalances in upstream data can induce persistent disparities that propagate to downstream tasks [10].

More recent work, however, challenges the relevancy of underrepresentation as an explanation for unfair outcomes. Dominguez-Catena et al. [9] show that models trained without exposure to a demographic group can still generalize to it with limited performance degradation. Ceccon et al. [4] find that combinations of label bias and proxy variables often exert a stronger influence on fairness than underrepresentation alone, while Legast et al. [16] analyze label bias and multiple forms of selection bias, demonstrating that their effects vary considerably depending on the dataset and evaluation setting.

A complementary line of work characterizes structural properties of the data that determine when underrepresentation is actually harmful. Jones et al. [15] study subgroup separability in medical image classification, showing that the degree to which models can distinguish demographic groups determines how dataset imbalance translates into performance disparities, producing either group-specific errors or more uniform degradation across groups. Maity et al. [19] demonstrate that cross-group generalization depends critically on how feature-label relationships vary across subpopulations, a factor that Wang et al. [26] independently identify in the context of intersectional

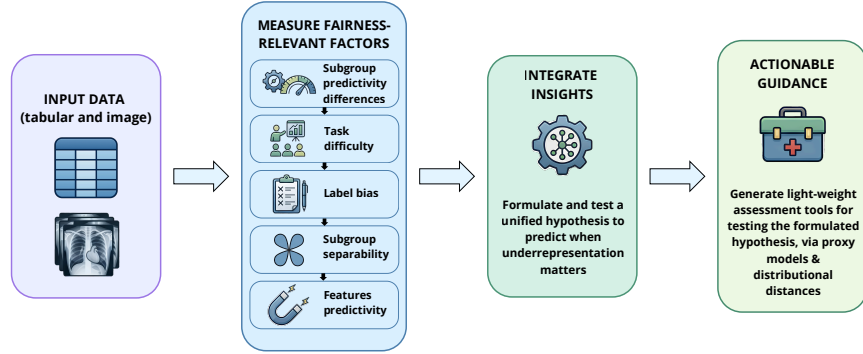


Fig. 1. Overview of the proposed pipeline. Starting from tabular and image datasets, the pipeline first measures fairness-relevant dataset characteristics. These are then integrated into a unified hypothesis predicting when underrepresentation constitutes a critical fairness risk. Finally, lightweight assessment tools based on proxy models and distributional distances are developed to enable practitioners to test this hypothesis directly from data, prior to model training.

fairness under the term *subgroup predictivity differences*. Claucich et al. [6] further show that disparities can arise from differences in inherent task difficulty across groups, and that enforcing balanced representations may be suboptimal when some groups are intrinsically harder to model. Similarly, Ceccon et al. [4] find that balanced representation can be detrimental in the presence of label bias. Finally, Legast et al. [16] highlight the role of features predictivity: when non-sensitive features carry little predictive signal, models rely more heavily on the sensitive attribute, amplifying the impact of underrepresentation.

Taken together, these findings paint a nuanced picture: fairness outcomes are shaped by structural properties of the data, of which underrepresentation is only one. Yet these factors have been studied in isolation, and no unified account exists of how they jointly determine whether imbalance translates into harmful disparities. Moreover, they are typically assessed only after training, leaving practitioners without practical guidance at the preparation stage.

3 Planned methodology

We sketch a methodological pipeline to systematically study the dataset characteristics that the literature has identified as influencing both model fairness and the impact of underrepresentation on the resulting model.

The factors we consider are:

- *subgroup predictivity differences*, whether the relationship between features and outcomes differs across groups [19, 27];
- *task difficulty difference*, how challenging it is for the model to predict each group [6];
- *label bias*, systematic noise in the labels of minority groups [4];
- *subgroup separability*, the degree to which sensitive information is encoded in the input features [15];
- *global feature predictivity*, the predictive power of non-sensitive features [16].

We additionally control for dataset size, base rate differences across groups, and whether the sensitive attribute is binary or multi-valued, as these may confound the analysis.

We plan to conduct experiments across tabular and image datasets. For image data, we consider medical imaging benchmarks central to fairness evaluation, including ChestX-ray14 [28] and Fitzpatrick17k [12], as well as CelebA [17], given the prominence of underrepresentation concerns in face recognition [3, 14]. For tabular data, we will consider the Student Performance dataset [8] and COMPAS [1], complemented by a synthetic dataset where the relevant characteristics can be directly controlled. We will measure fairness through a broader set of metrics, including performance-based group disparities, such as gaps in true positive rate, balanced accuracy, and AUC; outcome-based group fairness metrics, such as demographic parity; and individual fairness metrics, evaluated through consistency. We operationalize the impact of underrepresentation as the change in these metrics between a fully balanced setting and one where a minority group is entirely absent from training.

To isolate the contribution of individual factors, we plan to measure each one and, where possible, vary them independently through controlled manipulations: dataset size and base rates will be adjusted via subsampling; separability will be manipulated in tabular settings by removing features correlated with the sensitive attribute, and measured in image settings as the AUC of a classifier trained to predict it [4]; label bias will be introduced by randomly flipping a portion of positive labels within a minority group; and feature predictivity will be controlled by iteratively removing the most influential features as identified by SHAP [18]. Subgroup predictivity differences and task difficulty difference cannot be directly manipulated, and will instead be measured post-hoc: the former by comparing feature importance profiles across groups using SHAP [18], as a proxy for differences in the underlying feature-outcome relationships; the latter by training separate models on each subgroup and comparing their evaluation performance.

Once the most influential factors are identified and integrated into a unified hypothesis, we plan to develop lightweight ways to assess them prior to model training. This is particularly relevant in image settings where training full deep learning models is costly. We envision two complementary directions.

The first involves training smaller, faster proxy models for the prediction task that, due to their limited capacity, will not achieve optimal predictive performance, but can still reliably signal the presence of the relevant factors following the measurement procedures described above. The second direction exploits distributional distances. Most of the factors we consider can be interpreted as discrepancies between underlying distributions: subgroup separability, for instance, reflects a difference in the input distribution across groups, while subgroup predictivity, label bias, and task difficulty reflect differences in the conditional distribution of the target. For this reason, a measurement framework based on distributional distances seems a natural fit for many of the relevant factors. We therefore plan to investigate the use of such measures, taking inspiration from recent advancements in the context of fair representations [24]. Through this pipeline, we aim to characterize the conditions under which underrepresentation constitutes a critical fairness risk and those under which it can be safely disregarded, providing practitioners with principled guidance on when demographic imbalance requires active correction.

4 Conclusion

Determining whether demographic imbalance warrants intervention remains an open and practically consequential question. In this work, we have outlined a research agenda aimed at addressing it systematically, by identifying the dataset characteristics that govern the impact of underrepresentation on fairness and developing lightweight tools to assess them prior to training. We hope this pipeline will provide a principled foundation for future work on data-centric fairness, and invite discussion on its design and scope.

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